

Binary -- Guided Notes

Unit tp-1.2 | CompTIA Network+ N10-009 Obj. FC0-U71 1.2 | N10-008 FC0-U71 1.2

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

Binary is a base-_____ number system that uses only the digits _____ and _____. Each position in a binary number represents a power of two. The rightmost position represents $2^0 =$ _____, and the position immediately to its left represents $2^1 =$ _____.

Question 2

Fill in the 8 bit-position values from left (most significant) to right (least significant):

___ | ___ | ___ | ___ | ___ | ___ | ___ | ___

Question 3

Convert the binary number 10110010 to decimal. Show your work by listing which bit positions are set to 1 and adding their values.

Question 4

Convert the decimal number 205 to binary using the subtraction method. Start at 128 and work right. Show each step.

Question 5

One byte equals _____ bits. One nibble equals _____ bits. A byte can represent values from 0 to a maximum of _____. The maximum value occurs when all 8 bits are set to _____.

Question 6

The MSB stands for _____ and is the _____ (leftmost/rightmost) bit in a binary number. It has the _____ value. The LSB stands for _____ and is the _____ (leftmost/rightmost) bit, with value _____.

Question 7

Computers use binary because transistors -- the basic building blocks of processors -- have exactly _____ stable states: fully _____ (conducting, = 1) and fully _____ (not conducting, = 0). This makes binary arithmetic reliable and easy to implement in _____.

Question 8 -- Real World Application

REAL WORLD: A student claims that the binary number 11111111 is the same as the decimal number 255. Another student says it must be more because there are eight 1s. Explain who is correct and why, showing the math using bit position values.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/tp-1-2-binary/tp-1-2-binary-answer-key.pdf
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